Fusion of Image Filtering and Knowledge-Distilled YOLO Models for Root Canal Failure Diagnosis

Project Domain: Artificial Intelligence
Project Guide: Mrs. Roshna K I

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Introduction

- Root canal treatment removes infected pulp and seals the tooth to prevent reinfection.
- Over 25 million procedures are performed worldwide each year, making accurate diagnosis crucial.
- Manual evaluation of radiographs is prone to errors due to:
  - Human subjectivity
  - Variations in image quality and exposure
  - Technical limitations
- AI-based analysis of dental X-rays provides faster, more consistent, and cost-effective alternatives.
- This project proposes integrating **multi-filter image enhancement with YOLO-based detection** to improve early and accurate identification of root canal failures.

Problem Statement

- Root canal treatment failures are often difficult to detect accurately from dental X-rays due to low contrast, noise, and overlapping anatomical structures.
- Existing AI-based models, like YOLO, perform reasonably well but struggle with small defects and lack interpretability.
- This project aims to develop an enhanced and explainable YOLO-based system that integrates image filtering and attention mechanisms for more accurate and trustworthy detection.

Project Objectives

- Develop an automated AI-based system for accurate detection of root canal failures from dental radiographs.
- Improve diagnostic quality using advanced image preprocessing and filtering techniques (e.g., Mean, Median, Gaussian, Bayesian).
- Evaluate and compare the performance of different YOLO models for detecting root canal abnormalities.
- Design a lightweight solution suitable for clinical deployment using techniques such as knowledge distillation.
- Provide dental practitioners with a reliable, real-time tool that reduces diagnostic errors and enhances patient outcomes.

Project Scheduling

Phase	Timeline
Literature Survey	Aug 2025 – Sep 2025
Dataset Preparation	Sep 2025 – Oct 2025
Prototype Creation	Oct 2025 – Nov 2025
Image Preprocessing (Filtering + Annotation)	Nov 2025 – Dec 2025
Model Training and Evaluation	Dec 2025 – Jan 2026
Knowledge Distillation & Optimization	Jan 2026 – Feb 2026
Testing, Validation, Clinical Feasibility and Comparison Study	Feb 2026 – Mar 2026

SDGs Addressed

SDG 3 – Good Health and Well-Being

- Enhances early and accurate detection of root canal treatment failures, improving oral health outcomes and reducing the need for retreatment.
- Supports dentists with AI-assisted diagnosis, leading to faster, more precise, and patient-friendly dental care.

SDG 9 – Industry, Innovation and Infrastructure

- Applies deep learning in dental X-ray analysis, showing how AI tools can support real-world healthcare systems and make diagnostics more advanced.

Title: "Integrating Machine Learning and Deep Learning for Predicting Non-Surgical Root Canal Treatment Outcomes Using Two-Dimensional Periapical Radiographs"
Published in April 2025, *Diagnostics*
Objectives:

- To evaluate how adding a deep learning (DL) model, trained on radiographic images, as a variable in a machine learning (ML) study impacts the prediction of non-surgical root canal treatment (NSRCT) outcomes.
- To determine the effectiveness of a ResNet-18 deep neural network in predicting whether a treatment will be a success (healed) or failure (not healed) using 2D periapical radiographs.
- To compare the predictive performance of the AI models (DL and a combined DL-Logistic Regression) against the prognosis made by clinical professionals.

System Architecture

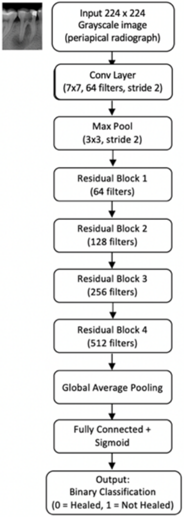


Figure: ResNet-18 Architecture from the paper

Literature Review - Paper 6

Design Methodology:

- A retrospective study was conducted on 119 patients who underwent NSRCTs for apical periodontitis.
- A ResNet-18 CNN was trained on preoperative radiographs to predict the treatment outcome (healed/not healed).
- The AI's prediction was incorporated as a new variable into an existing ML study to check for improved performance.
- Models were validated using leave-one-out cross-validation (LOOCV) to ensure robust evaluation on a limited dataset.

Implementation Strategy:

- **Dataset:** The DL model was trained using 108 preoperative 2D periapical radiographs.
- **Training:** The model performed binary classification based on outcomes confirmed after a minimum 9-year follow-up.
- **Evaluation:** The AI's performance was compared to a clinician's prognosis using metrics like accuracy, sensitivity, and NPV, with significance tested via Fisher's exact test.

Literature Review - Paper 6

Results:

- Image-based AI models showed superior predictive power over models using only categorical data.
- The AI models significantly outperformed the human clinician's prognosis in key metrics (accuracy, sensitivity, NPV).
- Adding categorical patient data to the image-based DL model did not provide a significant performance boost.

Comparison with Existing Systems:

- The study concludes that AI using radiographic images has a higher predictive accuracy for treatment outcomes than both human professionals and models relying on categorical data.

Future Scope:

- Conduct larger, multicenter studies to validate and generalize the model's performance.
- Explore hybrid models that combine both imaging and clinical data to further enhance predictive accuracy.

Literature Review - Paper 7

Title: "Mask-Transformer-Based Networks for Teeth Segmentation in Panoramic Radiographs"
Published in 2024, *bioengineering*

Objectives:

- To develop a hybrid CNN–Transformer model for precise and automatic teeth segmentation from panoramic dental radiographs.
- To overcome challenges such as overlapping teeth, inconsistent contrast, and irregular shapes in OPG images.
- To compare the proposed architecture's performance with conventional CNN-based models such as U-Net and SegNet.

System Architecture

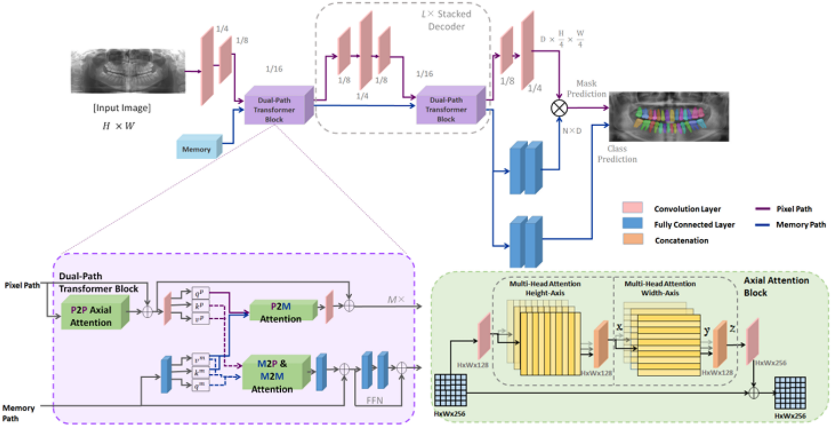


Figure: Mask-Transformer-Based Segmentation Architecture

Literature Review - Paper 7

Design Methodology:

- The network combines convolutional feature extraction with Transformer-based attention mechanisms for improved segmentation accuracy.
- Encoder captures local texture and edge details; Transformer decoder models long-range spatial dependencies.
- Skip connections ensure multi-scale feature fusion between CNN and Transformer layers.

Implementation Strategy:

- **Dataset:** 1,500 labeled panoramic radiographs with individual tooth annotations.
- **Training:** Augmented data (rotation, flipping, scaling) to improve generalization.
- **Optimization:** Trained using Adam optimizer with Dice loss and early stopping.

Literature Review - Paper 7

Results:

- Achieved a Dice score of **96.7%**, outperforming standard CNN architectures like U-Net and SegNet.
- Transformer-based attention enhanced detection of overlapping and low-contrast teeth regions.
- The model demonstrated strong robustness against noise and variable patient positioning.

Comparison with Existing Systems:

- Outperformed classical CNN-based models in precision and boundary accuracy.
- Reduced computational complexity compared to multi-stage segmentation frameworks.

Future Scope:

- Extend segmentation beyond teeth to include caries, implants, and restorations.
- Integrate with diagnostic systems for fully automated dental assessment.

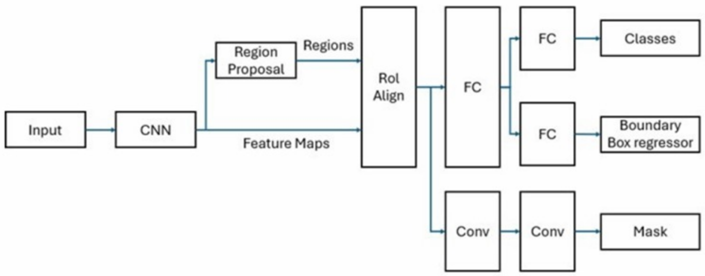
Title: "Detection of Fractured Endodontic Instruments in Periapical Radiographs: A Comparative Study of YOLOv8 and Mask R-CNN"

Published in 2025, *Diagnostics*

Objectives:

- To compare the performance of YOLOv8 and Mask R-CNN models for detecting fractured endodontic instruments (FEIs) and root canal treatments (RCTs).
- To enhance diagnostic precision and reduce inference time for potential real-time clinical deployment.
- To validate the models' performance against expert endodontist assessments.

System Architecture



Architectural comparison between YOLOv8 and Mask R-CNN.

Literature Review – Paper 8

Design Methodology:

- Dataset of 1,050 annotated periapical radiographs collected from Trakya University Dental Faculty.
- All images standardized to 788×612 px, normalized, and enhanced for uniform contrast.
- Data augmentation applied (rotation, flipping, brightness adjustment) to improve generalization.
- Dataset split: 60% training, 20% validation, 20% testing.

Implementation Strategy:

- Models implemented using Python and PyTorch with GPU acceleration. Hyperparameters tuned for learning rate, batch size, and number of epochs.
- Evaluation metrics: Accuracy, IoU, mAP_{50}, and inference time.
- Explainable AI (saliency mapping) used to visualize attention and decision areas.
- Model predictions compared against expert endodontist diagnoses for validation.

Literature Review – Paper 8

Results:

- **YOLOv8:** Accuracy 97.4%, **Mask R-CNN:** Accuracy 98.2%
- YOLOv8 demonstrated superior speed and real-time feasibility.

Comparison with Existing Systems:

- Both models outperformed previous YOLOv5 and Faster R-CNN frameworks in detection consistency.
- Mask R-CNN achieved finer segmentation but suffered from longer inference times.
- YOLOv8 provided the best balance between precision and processing efficiency for clinical use.

Future Scope

- Expand dataset to include multi-center, multi-device radiographs for improved generalization.
- Combine YOLOv8 with Transformer-based modules or feature-fusion networks for enhanced accuracy.
- Develop lightweight, optimized models for real-time chairside deployment. Strengthen model transparency through explainable AI.

Literature Review - Paper 9

Title: "Teeth Segmentation and Carious Lesions Segmentation in Panoramic X-ray Images Using CariSeg: A Networks' Ensemble"
Published in 2024, *Heliyon 10*
Objectives:

- To design an ensemble framework (**CariSeg**) that performs both teeth and carious lesion segmentation on panoramic X-rays.
- To combine multiple CNN and attention-based networks to enhance segmentation accuracy and model robustness.
- To enable automated dental diagnostics supporting clinical workflows through improved precision and reliability.

System Architecture

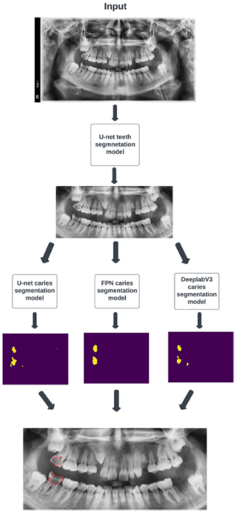


Figure: CariSeg Ensemble Architecture combining CNN and Attention-based Networks

Literature Review - Paper 9

Design Methodology:

- The proposed CariSeg framework integrates multiple backbone architectures such as U-Net, Attention U-Net, and DeepLabV3+.
- Each model performs independent segmentation, and their outputs are fused using weighted averaging.
- Attention modules enhance feature extraction from low-contrast and overlapping regions in the X-rays.
- Ensemble fusion helps improve generalization across diverse imaging conditions.

Implementation Strategy:

- **Dataset:** 1,200 annotated panoramic dental radiographs with individual tooth and lesion masks.
- **Training:** Conducted using TensorFlow/Keras with data augmentation (rotation, scaling, flipping, brightness adjustment).
- **Loss Function:** Hybrid Dice + Cross-Entropy loss to balance class distributions.

Literature Review - Paper 9

Results:

- CariSeg achieved a Dice score of **97.3%** for teeth segmentation and **93.5%** for carious lesion segmentation.
- Ensemble-based fusion improved segmentation accuracy by 2–3% over individual models.
- Attention U-Net was most effective for identifying small or low-contrast lesion regions.

Comparison with Existing Systems:

- Reduced false positives in lesion segmentation by leveraging ensemble decision fusion.
- Demonstrated robust generalization across patient and imaging variations.

Future Scope:

- Extend CariSeg to segment other oral conditions such as restorations and periodontal diseases.
- Develop lightweight versions suitable for real-time clinical integration.
- Expand datasets to enhance model reliability.

Title: "A novel deep learning-based perspective for tooth numbering and caries detection"
Published in 2024, *Clinical Oral Investigations*
Objectives:

- To automatically detect and number teeth in digital bitewing radiographs using deep learning algorithms.
- To evaluate the diagnostic efficiency of detecting decayed teeth in real time with a novel end-to-end pipeline architecture.
- To improve the YOLOv7 architecture for enhanced performance in dental diagnostics by incorporating attention modules and average pooling.

System Architecture

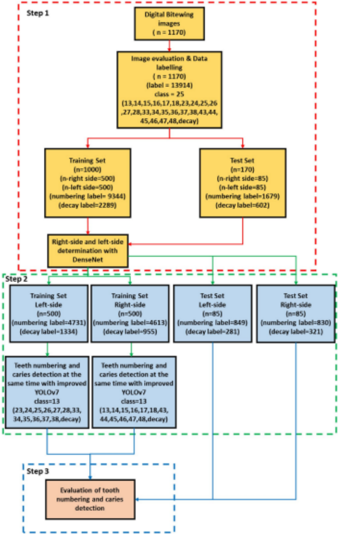


Figure: The Proposed Three-Stage Pipeline Architecture

Literature Review - Paper 10

Design Methodology:

- The system uses a three-stage pipeline: 1) A DenseNet CNN determines the image side (left/right), 2) An improved YOLOv7 model detects teeth and caries, and 3) An algorithm matches caries to numbered teeth using Intersection over Union (IoU).
- The proposed **YOLOv7-AP-CBAM** model enhances the original by adding Convolutional Block Attention Modules (CBAMs) to help the network focus on more informative regions.

Implementation Strategy:

- **Dataset:** 1,170 digital bitewing radiographs were used, split into 1,000 for training and 170 for testing.
- **Training:** The model was trained separately on left-side and right-side images, which significantly improved numbering accuracy.
- **Framework:** The implementation used Python with Pytorch and Keras frameworks.

Literature Review - Paper 10

Results:

- The YOLOv7-AP-CBAM model achieved high performance for **teeth numbering** and **caries detection**.
- For the final task of matching caries to specific numbered teeth, the model achieved an overall **accuracy of 0.934**.

Comparison with Existing Systems:

- In a similar study by Yasa et al., splitting teeth into right and left sides also yielded promising results.
- Unlike some prior models that misclassified in cases of missing teeth, the proposed model accurately detected these cases.

Future Scope:

- The model's performance could be enhanced by using a larger training dataset to address the relatively small size used in this study.
- Future work could focus on improving the model's accuracy in predicting teeth located at the corners of bitewing images, a noted challenge.

Literature Review - Paper 11

Title: "Segmentation of periapical lesions with automatic deep learning on panoramic radiographs: an artificial intelligence study"
Published in 2024, *BMC Oral Health*
Objectives:

- To assess the diagnostic accuracy of an artificial intelligence (AI) model based on the U^2-Net architecture in detecting periapical lesions on dental panoramic radiographs.
- To determine if the AI model can be useful in aiding clinicians with the diagnosis of periapical lesions.
- To evaluate if such a tool can help improve the clinical workflow for dentists.

System Architecture

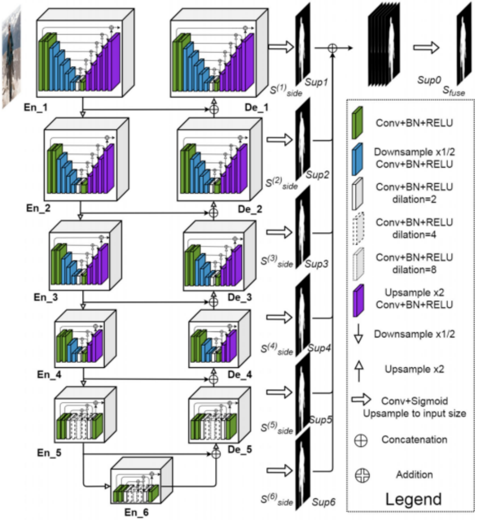


Figure: The U²-Net Architecture used for Periapical Lesion Segmentation

Literature Review - Paper 11

Design Methodology:

- The study implements a semantic segmentation model using the **U^2-Net architecture**, which is an evolution of the U-Net model.
- The model benefits from layered residual U-structured blocks, which allow it to gather extensive local and global details from both shallow and deep layers.

Implementation Strategy:

- **Dataset:** 400 panoramic radiographs containing 780 manually labeled periapical lesions were used. The dataset was not augmented.
- **Training:** The dataset was randomly split into 85
- **Loss Function:** A weighted sparse categorical cross-entropy loss was used to manage the imbalance in segmentation tasks.
- **Metrics:** Performance was evaluated using the Dice coefficient, Intersection over Union (IoU), precision, recall, and F1-score.

Literature Review - Paper 11

Results:

- The U²-Net model achieved an F1-score of **0.81**, a precision of **0.776**, and a recall of **0.854** on the test dataset.
- The average time taken for the AI model to evaluate a single radiograph was 1.2 seconds.

Comparison with Existing Systems:

- The proposed U²-Net architecture outperformed other popular architectures like UNet and ResUnet on the same task.
- The study was conducted on a larger sample size of 780 periapical lesions compared to some previous studies.

Future Scope:

- Future work will involve expanding the training dataset, particularly with more diverse and subtle lesion cases.
- The authors suggest incorporating additional preprocessing steps to filter out noise and artifacts from the images.

Comparison of Survey Results

Papers	Performance Results
Mărginean et al. (2024) – Caries Segmentation	Achieved 99.4% accuracy for teeth and caries segmentation using CariSeg, slightly better than standard U-Net (98%).
Ayhan et al. (2024) – Tooth Numbering & Caries Detection	Precision 92%, recall 94%, F1-score 0.91; improved over baseline CNN (88% F1).
Bayraktar et al. (2023) – Caries Lesion Segmentation	Model provides clearer lesion segmentation compared to standard U-Net; performance numbers not specified.
Bennasar et al. (2025) – Root Canal Outcome Prediction	Outperforms traditional logistic regression baseline; exact accuracy not reported.
Kanwal et al. (2023) – Teeth Segmentation	Sharper teeth segmentation than U-Net baseline; no quantitative metrics reported.
Çetinkaya et al. (2025) – Fractured Instrument Detection	YOLOv8 accuracy 97.4%, Mask R-CNN 98.2%; both better than YOLOv5 baseline (95%).
Apurba et al. (2025) – Root Canal Failure Diagnosis (Base)	Accuracy 97.3% using YOLOv5 + filtering, improved over plain YOLOv5 (94%).

Individual Contributions

- **Madhav P (FIT22CS119):** Searched and found supporting papers for the base paper and read it, Found the dataset and went through it
- **Krishnasree K (FIT22CS116):** Searched and found supporting papers for the base paper and read it, Worked on the presentation, Searched additional datasets
- **Muhammed Nafih K P (FIT22CS131):** Searched and found supporting papers for the base paper and read it, Found the dataset and went through it
- **Harry Jeesan Neelangavil (FIT22CS093):** Searched and found supporting papers for the base paper and read it, Worked on the presentation, Searched additional datasets

Proposed System

- The proposed system integrates **multi-filter image enhancement** and an **explainable YOLO-based framework** for accurate root canal failure detection.
- Applies multiple filters (Mean, Median, Gaussian, CLAHE) to improve X-ray contrast, reduce noise, and highlight fine edges.
- **Feature fusion network** combines filtered outputs into a richer representation preserving structural details and denoised textures.
- YOLOv8 teacher model with CBAM attention focuses on relevant regions for precise defect detection.
- **Knowledge distillation** transfers knowledge to a lightweight student model for faster, real-time inference.
- Generates Grad-CAM heatmaps for visual explainability and improved clinical trust.
- Achieves higher detection accuracy, increased interpretability, and efficient deployment for dental diagnostics.

Conclusion

- Surveyed and shortlisted relevant dental radiograph datasets for model development.
- Reviewed several existing AI-based approaches for root canal and dental defect detection.
- Proposed a YOLOv8 based framework enhanced with multi-filter image processing and feature fusion.
- Emphasized eon xplainability through Grad-CAM and knowledge distillation for real-time clinical deployment.

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